Q6-For the reaction: $CaCl_{2(aq)} + Na_2CO_{3(aq)} \rightarrow CaCO_{3(s)} + 2NaCl_{(aq)}$. If 2.50 g of $CaCl_2$ are completely dissolved in a beaker of water and 2.50 g of Na_2CO_3 are completely dissolved in water in a second beaker. The two solutions are mixed and are allowed to react to form a $CaCO_3$ precipitate and aqueous NaCl solution. *Molar masses: Na_2CO_3=106.00 g/mol., $CaCl_2$= 111.00 g/mol.*

I. In this experiment Na_2CO_3 is the limiting reactant.

II. The molar mass of Na_2CO_3 is less than the molar mass of $CaCl_2$.

A) Statement I is true, Statement II is true
B) Statement I is true, Statement II is false
C) Statement I is false, Statement II is true
D) Statement I is false, Statement II is false
E) None of the above

Q7- What is the percent yield of the following reaction: $CaCO_{3(s)} \rightarrow CaO_{(s)} + CO_{2(g)}$, if 60 grams of $CaCO_3$ are heated to give 15 grams of CaO? *Molar masses: $CaCO_3$= 100.00 g/mol., CaO = 56.08 g/mol.*

A) 32 %	B) 45 %	C) 60 %	D) 74 %	E) 83 %

Q8- How many moles of gas are found in a 50.0 L container at 22.0°C and 825 torr?

A) 2.29×10^4	B) 1.70×10^3	C) 2.23	D) 0.603	E) 18.4

Q9- What is the empirical formula for a compound containing: 1.3 mol carbon and 5.2 mol hydrogen.(At.wts in g/mol: C=12,H=1)

A) CH	B) CH_4	C) C_2H_4	D) C_2H_4	E) C_3H_8

Q10- During empirical formula experiment, what happens when a few droplets of water are added to magnesium nitride?

A) NH_3 gas is released
B) magnesium nitride is converted to H_2O and magnesium oxide
C) N_2 gas is evolved
D) H_2 gas is evolved
E) None of the above

Q11- A hydrated salt ($LiCl.xH_2O$) is heated in a crucible until all water is driven off. If mass percent of water =29.7%.The molar mass (in g/mol) of the hydrated salt equals _____. (**At.wts**: Li =7.0, H=1.0, O=16.0 Cl=35.5 g/mol)

A) 60.5	B) 78.5	C) 270.3	D) 161	E) 216.8

Q12- The boiling point of a liquid depends on _____

A) Strength of intermolecular forces
B) External atmospheric pressure
C) molecular weight of the liquid

Use the following data when needed R=0.08206L.atm/K.mol , 1atm=760torr=760mmHg

Atomic masses (g/mol) = H:1 ,O:16 ,C:12 ,N:14 , Cl:35.5 ,S:32 ,Ca:40

PART1 Choose the answer of the following question's and transfer it in CAPITAL letters to table

Q1-A sample of calcium carbonate **$CaCO_3$ (s)** absorbs 45.5 J of heat, and its temperature increases from 21.1 °C to 28.5 °C. If the specific heat of calcium carbonate is 0.82 J/g-K, what is the mass (in grams) of the sample?

A) 3.7	B) 5.0	C) 7.5	D) 4.1	E) 5.0×10^3

Q2- The enthalpy for dissolution (dissolving a solid in water) of solid NaOH equals 44.4 kJ/mol. Calculate the maximum temperature (in °C) measured by a thermometer when 10.1g sample of NaOH dissolves in 250.0 g of water in a coffee-cup calorimeter, the initial temperature of water before adding NaOH = 23.0 °C . (Assume specific heat of NaOH solution= 4.18 J/g.°C.)

A) 35.2	B) 24.0	C) 34.7	D) 33.3	E) 40.2

Q3- The mass percent of NaOCl (molar mass =74.5 g/mol) in a bleach sample = 5.5%. A 2.5 mL sample of bleach is diluted to 25.0mL and titrated with a standard solution of $Na_2S_2O_3$. The sample required 8.4 mL of $Na_2S_2O_3$ to reach the end point. If the density of the original bleach solution =1.04g/mL.Calculate the molarity of $Na_2S_2O_3$ solution.

A) 0.77 M	B) 0.68M	C) 0.37M	D) 0.55 M	E) 0.46M

Q4- Which of the following will end up with an **incorrect** measured value of percentage of NaOCl during analysis of a bleach sample being titrated with $Na_2S_2O_3$

A) Washing the inside walls of the flask during the titration
B) Dilution of the original bleach sample.
C) Adding starch to the bleach solution before starting titration with $Na_2S_2O_3$
D) The appearance of a blue color when starch indicator is added.
E) Reading V_f of the burette when color changes from blue to white (colorless).

Q5- Avinegar sample is titrated a 0.2M $Ba(OH)_2$ according to the following equation:

Use data given in table below to find the initial burette reading (Vi)

$$2CH_3COOH + Ba(OH)_2 \longrightarrow 2\,H_2O + Ba(CH_3CO_2)_2$$

A) 6 mL
B) 17 mL
C) 11 mL
D) 22 mL
E) 25 mL

Mass of empty flask	**25.7 g**
Mass of flask + vinegar sample	**30.5 g**
% CH_3COOH in sample	**5.5%**
Mm of CH_3COOH	**60.0 g/mol**
The final burette reading	**28 mL**

D) All of the above
E) None of the above

Q13- Which of the following is a soluble silver (Ag) salt in water :

A) AgCl	B) Ag_2CO_3	C) $AgNO_3$	D) Ag_2SO_4	E) All are soluble

PART2

Answer the following question's and *transfer your answer to the table in the cover page*

Q1 Given the following reactions

$Fe_2O_3\ (s) + 3CO\ (s) \rightarrow 2Fe\ (s) + 3CO_2\ (g)$ $\quad \Delta H = -28.0$ kJ

$3Fe\ (s) + 4CO_2\ (s) \rightarrow 4CO\ (g) + Fe_3O_4\ (s)$ $\quad \Delta H = -12.5$ kJ

the enthalpy of the reaction of Fe_2O_3 with CO

$3\ Fe_2O_3\ (s) + CO\ (g) \rightarrow CO_2\ (g) + 2\ Fe_3O_4\ (s)$ is ____-109_____ kJ.

Q2- For the equation: $OCl^- + 2I^- + 2H^+ \rightarrow Cl^- + I_2 + H_2O$ .The reducing agent is _____I-____

Q3- The indicator used in vinegar analysis experiment is_____ Phenolphthalein

Q4- If by mistake, 40 mL of H_2O are added to a vinegar sample instead of 20 mL. How is the calculated percentage of acetic acid (% CH_3COOH) affected by this mistake?_____ not affected/ the same

Q5- The Solid material formed in the reaction $Na_2CO_3 + CaCl_2 \rightarrow CaCO_{3\,(S)} + 2NaCl$ was (experimentalsteps) _____ Filtered, washed with cold water, and then allowed to dry.

Q6- Suppose you want to determine the molar mass of two liquids, **toluene (B.p 110°C)** & **dichloromethane (B.p 40°C)**. Which liquid's molar mass could be determined by Dumas method?

_____

Q7- The two compounds formed when magnesium turnings are heated in a crucible and exposed to air are:

_____

_____

Q8- Write the general formula of alums. _____

Q9- If a piece of wood contaminates a liquid, this cause the boiling point of liquid to _____

Q10- A block of iron (mass= 1105 g) has dimensions of 4.5 cm, 5.2 cm and 6 cm
Calculate density of iron

